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Abstract	Reliable global visual localization of unmanned aerial vehicles (UAVs) in GNSS-denied environments remains vulnerable to fog, snow, and nighttime domain shifts, while generative augmentation can invalidate pixel-level labels by altering object geometry. In this work, we propose Geo-consistent Weather Augmentation (GCWA), a fail-closed Generate–Verify–Filter framework that accepts image-to-image weather transformations only when mask intersection over union (IoU) and the structural similarity index measure (SSIM) applied to edge maps (Edge-SSIM) preserve building geometry and a perceptual-shift criterion confirms a nontrivial weather change. The accepted samples are used to fine-tune the YOLOv11 model and construct a compact structural descriptor for aerial visual place recognition (VPR). In cross-domain segmentation, baseline training degraded from 0.826 mean average precision at an IoU threshold of 0.50 (mAP50) on original images to 0.397 in fog, whereas GCWA mixed-domain training achieved 0.824, 0.815, 0.724, and 0.753 mAP50 on original, fog, night, and snow tests, respectively. On the VPAIR urban subset, YOLOv11+GCWA obtained top-1 recall (Recall@1) values of 0.121, 0.139, and 0.125 in fog, night, and snow and retained 65.8% of clean-condition performance, compared with 46.6% for the same YOLOv11 descriptor without GCWA and 39.4% for MixVPR; MixVPR remained best in clean conditions at 0.210 versus 0.195. Verification required 33.07 ms per candidate, and online segmentation averaged 6.54 ms on the tested desktop graphics processing unit. The results demonstrate that verified geometric consistency is an effective training-data constraint for adverse-weather UAV localization, while real-weather and embedded-platform validation remain necessary.	
Keywords (separated by '-')	VPR - UAV - YOLO - Weather augmentation - Domain shift - Segmentation	
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A Geo-consistent Weather Augmentation Method for Improving UAV Localization Robustness under Adverse Meteorological Conditions

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Abstract

Reliable global visual localization of unmanned aerial vehicles (UAVs) in GNSS-denied environments remains vulnerable to fog, snow, and nighttime domain shifts, while generative augmentation can invalidate pixel-level labels by altering object geometry. In this work, we propose Geo-consistent Weather Augmentation (GCWA), a fail-closed Generate–Verify–Filter framework that accepts image-to-image weather transformations only when mask intersection over union (IoU) and the structural similarity index measure (SSIM) applied to edge maps (Edge-SSIM) preserve building geometry and a perceptual-shift criterion confirms a nontrivial weather change. The accepted samples are used to fine-tune the YOLOv11 model and construct a compact structural descriptor for aerial visual place recognition (VPR). In cross-domain segmentation, baseline training degraded from 0.826 mean average precision at an IoU threshold of 0.50 (mAP50) on original images to 0.397 in fog, whereas GCWA mixed-domain training achieved 0.824, 0.815, 0.724, and 0.753 mAP50 on original, fog, night, and snow tests, respectively. On the VPAIR urban subset, YOLOv11+GCWA obtained top-1 recall (Recall@1) values of 0.121, 0.139, and 0.125 in fog, night, and snow and retained 65.8% of clean-condition performance, compared with 46.6% for the same YOLOv11 descriptor without GCWA and 39.4% for MixVPR; MixVPR remained best in clean conditions at 0.210 versus 0.195. Verification required 33.07 ms per candidate, and online segmentation averaged 6.54 ms on the tested desktop graphics processing unit. The results demonstrate that verified geometric consistency is an effective training-data constraint for adverse-weather UAV localization, while real-weather and embedded-platform validation remain necessary.

Keywords VPR · UAV · YOLO · Weather augmentation · Domain shift · Segmentation

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1 Introduction

Reliable global localization is a prerequisite for autonomous UAV operation when global navigation satellite system (GNSS) measurements are unavailable, degraded, or denied. VPR supports this requirement by retrieving reference imagery that depicts the same place as a query image. Its reliability, however, depends on retaining discriminative evidence under simultaneous changes in viewpoint, scale, illumination, season, and scene appearance [1]. Deep representations have improved retrieval accuracy, but perceptual aliasing, dataset bias, and severe domain shifts remain persistent causes of false or missed matches [2]. General place recognition therefore requires not only high in-domain accuracy, but also transfer across platforms, environments, and operating periods [3].

Cross-view UAV localization is especially difficult because downward-looking UAV observations must be matched to georeferenced satellite images acquired with different sensors, resolutions, altitudes, and illumination. Attention-based cross-view models can learn relations between the two image domains [4], while aerial benchmarks show that large-scale retrieval remains difficult even when both views are nominally nadir-oriented [5]. Urban semantics provide a complementary source of invariance because roofs, footprints, and spatial arrangements are generally more persistent than weather-dependent texture. Unsupervised landmark extraction from Earth imagery has demonstrated that intermediate segmentation features can identify distinctive structural regions [6]. Building masks and cross-view color normalization have likewise supported compact satellite-UAV descriptors [7].

The unresolved problem concerns the weather dependence of the features from which such descriptors are formed. Conventional augmentation broadens the empirical training distribution, but its validity remains task-specific and it does not guarantee that dense labels survive a transformation [8]. Generative augmentation can synthesize realistic adverse conditions and increase contextual diversity [9]; nevertheless, meteorological plausibility does not ensure geometric fidelity. A generated fog, snow, or night image may contain shifted roof edges, modified footprints, removed buildings, or hallucinated structures. Pairing such an image with the original mask introduces label noise precisely in the contours intended to encode stable urban geometry. Structure-preserving translation reduces this risk at the model level [10], but average structural regularization does not certify each generated training pair. Adverse-weather synthesis must therefore be treated as controlled dataset construction: a sample should be accepted only when the requested photometric shift is nontrivial and its inherited geometry remains valid.

The goal of this study is to improve UAV global localization robustness under fog, snow, and night conditions by learning weather-invariant, building-based visual descriptors from geometrically verified synthetic data. The objective of the study is to develop and evaluate GCWA, which generates adverse-weather images, rejects geometrically invalid candidates, and quantifies the resulting effects on cross-domain building segmentation and aerial VPR.

The major contributions are as follows:

- A GCWA formulation that converts adverse-weather synthesis into a Generate–Verify–Filter procedure and uses the original building annotations as explicit spatial constraints during image-to-image transformation;
- A fail-closed, multi-criterion acceptance rule combining Mask IoU, Edge-SSIM, and a minimum Learned Perceptual Image Patch Similarity (LPIPS) weather-shift

threshold, thereby excluding candidates that are geometrically inconsistent or photometrically ineffective;

- A leakage-free cross-weather protocol that separates original train and test partitions before augmentation and evaluates original, fog, snow, night, and mixed-domain training through a complete train-domain–test-domain transfer matrix;
- A segmentation-driven VPR pipeline that derives 256-dimensional descriptors from the You Only Look Once (YOLO) version 11 nano segmentation model (YOLOv11n-seg) and evaluates them on the urban Aerial Visual Place Recognition and Localization in Large-scale Outdoor Environments (VPAIR) subset against CosPlace, Feature Mixing for Visual Place Recognition (MixVPR), and a non-GCWA YOLOv11 baseline under a fixed clean satellite database.

The remainder of this paper is organized as follows. Section 2 reviews VPR under environmental change, generative augmentation, geometric integrity, and segmentation-based descriptors. Section 3 defines GCWA and the experimental protocol. Section 4 reports the segmentation and localization results. Section 5 interprets the findings and limitations, and Section 6 presents the conclusions.

2 Literature Review

2.1 VPR Challenges in Dynamic Environments

VPR must suppress nuisance variation while distinguishing different places. Deep-VPR surveys identify appearance and viewpoint change, perceptual aliasing, dataset bias, and scalability as persistent limitations [2]. General place-recognition research also highlights cross-platform transfer, efficiency, uncertainty, and generalization to unseen environments [3]. UAV imagery adds variation in altitude, orientation, resolution, and sensor radiometry.

Cross-view methods reduce discrepancies between UAV and satellite observations. The Single-Stream Pyramid Transformer (SSPT) learns spatial and cross-view interactions through self-attention and cross-attention [4]. It reduces viewpoint discrepancy but does not enforce invariance to fog, snow, or low illumination. Hierarchical detection combines context and fine-grained attributes to distinguish similar UAV targets [11]. However, it addresses target identification rather than large-scale geo-localization. VPAIR provides long aerial trajectories, satellite renders, and difficult in-plane rotations for UAV-query retrieval [5]. It supports cross-view evaluation but lacks adverse-weather image–mask pairs.

Long-term localization shows that the same place changes across seasons and times. The Robot Outdoor Visual Simultaneous Localization and Mapping (SLAM) Dataset for

Environmental Robustness (ROVER) exposes localization failures under low light and vegetation change [12]. iSimLoc uses simulated imagery and hierarchical matching in unseen environments [13]. This indicates that synthetic evidence can extend coverage when real observations are unavailable. Model-free re-ranking improves global-candidate ordering with local features without retraining the descriptor [14]. Yet it cannot restore scene structure already lost from the query representation. The Visual Place Recognition Benchmark (VPR-Bench) shows that no descriptor dominates across all viewpoint and appearance changes [15]. These findings favor training-time invariance over retrieval-stage correction alone.

Robotic inspection studies report related perception constraints but different outputs. Tunnel robots integrate locomotion, sensing, and communication in confined infrastructure [16]. Cable-tunnel enhancement compensates for low illumination, haze, dust, noise, and low contrast [17]. Blade-inspection robots reduce risks and access constraints by carrying sensors across large surfaces [18]. Collaborative aerial inspection introduces human assistance when automatic interpretation is uncertain [19]. These systems confirm the operational importance of degraded visibility, but they do not formulate adverse-weather generation for cross-view VPR. Multispectral fusion combines thermal and red–green–blue (RGB) blade imagery to recover complementary evidence [20]. However, it increases sensing and calibration demands. This study instead uses RGB imagery and verified structural augmentation.

2.2 Generative Models in Data Augmentation for Computer Vision

Conventional augmentation applies geometric, photometric, filtering, mixing, or erasing transformations. Its suitability depends on whether target semantics and annotation geometry remain valid [8]. Generative methods provide broader domain variation. A language–diffusion–segmentation pipeline diversifies backgrounds while protecting salient foregrounds [9]. Its semantic guidance benefits classification, although plausibility is weaker than pixel-level certification of inherited masks.

Adverse weather can be generated through adversarial or diffusion translation. A multi-weather generative adversarial network (GAN) generates and removes haze, rain, snow, and related conditions from unpaired road imagery [21]. It broadens domain coverage but may alter local texture and geometry. Weather-Controllable Diffusion-assisted Unsupervised Domain Adaptation (W-ControlUDA) uses controllable diffusion and filters unreliable pseudo-labels for semantic segmentation [22]. However, it filters adaptation labels rather than verifying source-boundary preservation

in every translated image. WeatherFLUX translates among sunny, rain, snow, fog, and night domains using weak pairing and semantic and frequency-aware constraints [23]. These constraints promote alignment but cannot guarantee pixel-wise agreement with the original mask.

Language and semantic conditioning improve control but not boundary fidelity. Data Augmentation Leveraging Diffusion Model and Large Language Model (LLM) with Adaptive Guidance Scaling (DALDA) enriches prompts and adjusts diffusion guidance using image–text consistency [24]. Semantic-guided diffusion uses class labels and generated descriptions to preserve category relevance [25]. Both optimize semantic correspondence rather than contours. An instance-level sim-to-real pipeline combines segmentation with Cycle-Consistent Generative Adversarial Network (CycleGAN) translation for robot-navigation pose estimation [26]. It reduces foreground corruption but is not a per-candidate verifier for dense masks. The Adverse-Weather Data Augmentation Network (AWDAN) augments adverse-weather data for traffic-object detection [27]. Detection boxes tolerate contour shifts that invalidate instance-segmentation labels.

Gemini 2.5 Flash Image accepts image-and-text inputs and supports editing and generation [28]. This enables controlled fog, snow, and night prompts while retaining the source image as context. However, it provides no geometric guarantee. Valid generation therefore requires both a visible weather shift and preserved building regions and contours. GCWA verifies these conditions before admitting candidates to training. Table 1 compares this requirement with major augmentation families.

2.3 Geometric Integrity Issues in Diffusion and Generative Models

For dense segmentation, generated imagery is useful only if inherited annotations remain spatially valid. Semantic masking for localization emphasizes persistent regions and compatible semantic classes [29]. Semantic visual SLAM uses object identity, boundaries, and spatial relations in complex scenes [30]. Moving a roof edge or footprint therefore changes the target even when the scene remains recognizable.

Structure-preserving translation separates or regularizes structural and textural information to change appearance without uncontrolled deformation [10]. Topology-preserving segmentation shows that similar overlap scores can hide connectivity and deformation errors [31]. Location-aware self-supervision preserves relative patch positions in dense features [32]. However, it learns representations rather than validating synthetic image–mask pairs. These approaches encourage geometric control but provide average behavior, not a fail-closed decision for each candidate.

Table 1 Comparison of data augmentation methods for computer vision

Method	Model type	Use in augmentation	Advantages	Limitations
GAN-based weather translation [21]	Adversarial	Weather simulation and removal	Realistic domain conversion from unpaired data	Training instability and possible structural drift
Controllable weather diffusion [22]	Diffusion-based	Weather-conditioned domain adaptation	Explicit weather control and diverse synthesis	High computational cost; no inherited-mask certificate
LLM-guided diffusion [24]	Hybrid	Context-aware prompt construction	Diverse, semantically relevant samples	Validity depends on prompt and guidance quality
Weakly paired image translation [23]	Translation-based	Multi-weather conversion	Broad weather coverage with limited pairing	Fine geometry may remain insufficiently constrained

A practical verifier should combine complementary criteria. Region overlap tests annotation agreement, edge similarity detects contour displacement, and perceptual distance rejects outputs lacking the requested weather change. Multi-metric evaluation is preferable to a single summary measure [33]. GCWA therefore accepts candidates only when geometric fidelity and nontrivial photometric change are jointly demonstrated.

2.4 Robust Descriptors Based on Semantic Segmentation

Map-representation surveys compare geotagged frames, landmarks, point clouds, vector semantic maps, and neural maps by compactness, invariance, and geometric precision [34]. Discriminative feature selection estimates regional stability and suppresses transient content during localization [35]. Progressive self-distillation transfers segmentation knowledge from normal to adverse weather without retaining source data [36]. These methods improve robustness but do not verify alignment between synthetic weather images and inherited annotations.

Semantic segmentation also supports aerial autonomy outside VPR. The Dynamic Trajectory Adaptation Method (DyTAM) segments turbine components and estimates blade orientation for adaptive UAV trajectories [37]. YOLO ensembles fuse visible and thermal detections for defect inspection [38]. These studies show the value of structural perception, but their outputs and acquisition geometries differ from satellite–UAV retrieval.

Building segmentation suits urban aerial localization because footprints, roof contours, and layouts are more persistent than texture. Intermediate YOLOv11 segmentation embeddings support unsupervised landmark outlier detection in Earth imagery [6]. A related cross-view method combines YOLOv11 building segmentation, color normalization, and satellite retrieval from UAV queries [7]. YOLOv11 offers instance-segmentation variants and efficient feature extrac-

tion across scales [39]. The Buildings dataset provides 9,665 annotated images for segmenter development [40]. The urban VPAIR subset supplies georeferenced UAV queries and satellite references [5]. This separation allows structural learning independently of the VPR benchmark.

Holistic VPR methods provide strong controls. CosPlace casts large-scale geo-localization as classification over geographic groups and produces compact descriptors [41]. MixVPR mixes spatial and channel information and balances accuracy with latency [42]. Neither explicitly anchors descriptors to building masks. Their comparison with the proposed vector tests whether verified structural training improves robustness beyond global aggregation.

Other methods introduce structure during aggregation or inference. Multi-Scale Fusion of Locally-Global Descriptors (Patch-NetVLAD) derives multi-scale patch descriptors and performs spatially informed matching [43]. AnyLoc aggregates self-distillation with no labels, version 2 (DINOv2) features using Vector of Locally Aggregated Descriptors (VLAD) and transfers across diverse environments without VPR-specific fine-tuning [44]. The Sinkhorn Algorithm for Locally Aggregated Descriptors (SALAD) uses optimal-transport aggregation and a dustbin cluster to discard uninformative features [45]. Segment-based Vector of Locally Aggregated Descriptors (SegVLAD) retrieves open-set semantic segment neighborhoods and reduces non-overlapping-region influence [46]. The backbone abbreviations used below are convolutional neural network (CNN), Vision Transformer (ViT), and Segment Anything Model (SAM). These architectures generally assume valid upstream training or adaptation images. GCWA instead controls generated image–mask admissibility before descriptor learning. Table 2 summarizes the distinction.

2.5 Summary and Identified Gaps

The literature covers cross-view representation learning, retrieval-stage reasoning, adverse-weather generation, and

Table 2 Comparison of visual place recognition approaches by source of structural information

Method	Backbone	Semantic anchoring	Geometric constraint	Descriptor per image
CosPlace [41]	CNN	None	None	Single global vector
MixVPR [42]	CNN	None	None	Single global vector
Patch-NetVLAD [43]	CNN	None	Spatial verification at inference	Multi-scale patch set
SALAD [45]	ViT (DINOv2)	Emergent, not explicit	None	Single global vector
AnyLoc [44]	ViT-G/14 (DINOv2)	Emergent, not explicit	None	49,152-D vector
SegVLAD [46]	ViT (SAM + DINOv2)	Explicit, open-set segments	Segment adjacency	Multiple segment vectors
GCWA (ours)	CNN (YOLOv11n-seg)	Explicit, building masks	Verified at dataset construction	256-D vector

robust semantic segmentation, but their intersection remains incomplete. VPR usually assumes valid upstream images and labels. Generative augmentation instead emphasizes prompt compliance, realism, alignment, or average downstream performance. None prevents an individual weather-translated image from retaining an invalid building mask.

Four omissions follow:

- Unverified geometric fidelity. Existing adverse-weather generators do not jointly test building overlap and contour preservation before admitting synthetic samples to dense segmentation.
- Absence of a fail-closed curation rule. Semantic guidance, cycle consistency, and pseudo-label filtering encourage average validity but do not reject every corrupted candidate.
- Limited linkage between segmentation robustness and VPR. Weather-robust segmentation and VPR descriptors are usually evaluated separately, obscuring the retrieval effect of mask-consistent synthetic training.
- Incomplete cross-domain evidence. Mixed-domain results cannot reveal asymmetric transfer among original, fog, snow, and night conditions or separate clean-domain costs from adverse-domain gains.

The contributions address these gaps at data, model, and evaluation levels. Generate–Verify–Filter treats augmentation as dataset construction rather than unconditional expansion. Mask IoU measures building-region consistency, Edge-SSIM measures contour stability, and minimum LPIPS confirms nontrivial appearance change. Failure of any criterion excludes the sample. A complete train-domain–test-domain matrix measures asymmetric transfer and clean-condition trade-offs. A fixed clean satellite database isolates robustness to weather degradation in UAV queries. Comparisons with CosPlace, MixVPR, and a non-GCWA YOLOv11 descriptor connect curation, segmentation, descriptor formation, and retrieval.

Accordingly, the study addresses the following tasks:

1. Quantify the effects of fog, snow, and night shifts on building segmentation and segmentation-derived feature stability;
2. Implement GCWA as a Generate–Verify–Filter process with a fail-closed acceptance decision;
3. Construct leakage-free original and adverse-weather train/test partitions using fixed Mask IoU, Edge-SSIM, and LPIPS thresholds;
4. Evaluate YOLOv11n-seg across all train-domain–test-domain combinations using mAP50 and the harmonic mean of precision and recall (F1 score);
5. Evaluate Recall@1 on the urban VPAIR subset against holistic and non-GCWA baselines while perturbing only UAV queries.

The study does not replace existing VPR descriptors or generators. It supplies quality control for adverse-weather images with inherited dense annotations and tests whether this layer improves segmentation and satellite–UAV localization.

3 Materials and Methods

This section defines the proposed GCWA framework and the experimental protocol used to evaluate it.

3.1 Mathematical Formalization of the Approach

The proposed localization strategy assumes that urban geometry, particularly building footprints, roof contours, and their spatial arrangement, provides a more persistent descriptive basis than weather-dependent appearance. Let $\mathcal{Q} = \{q\}$ denote the set of UAV query images and $\mathcal{R} = \{r_1, \dots, r_n\}$ the database of georeferenced satellite images. The localization model learns a mapping $\mathcal{F} : \mathcal{I} \rightarrow \mathcal{V}$ from the image space $\mathcal{I}$ to a descriptor space $\mathcal{V} \subset \mathbb{R}^d$.

The mapping $\mathcal{F}(\cdot)$ is implemented with a YOLOv11 segmentation model, whose recent variants have been applied

effectively in UAV visual tasks ranging from target localization to multispectral defect detection [38]. For each image, the feature tensor from the final convolutional layer preceding the segmentation head is globally pooled to obtain a fixed-length scene descriptor of dimension $d = 256$. Because the network is optimized to delineate buildings, this representation encodes structural regularities in built-up areas while reducing dependence on local color and texture. No additional descriptor normalization is applied. Retrieval therefore uses Euclidean distance, which preserves both feature direction and magnitude and provides a direct nearest-neighbor rule in $\mathcal{V}$.

For a query q , the index of the nearest satellite reference is

$$k = \arg \min_{1 \leq j \leq n} \|\mathcal{F}(q) - \mathcal{F}(r_j)\|_2, \quad (1)$$

where r_j is the j -th reference image and n is the reference-set size; Eq. 1 produces the top-ranked satellite candidate r_k .

Let $\ell : \mathcal{I} \rightarrow \mathbb{R}^2$ map an image to its georeferenced ground-truth coordinates, and let $\text{dist}(\cdot, \cdot)$ denote the distance between two coordinate pairs. A retrieval is correct when

$$\text{dist}(\ell(q), \ell(r_k)) \leq \tau_{\text{dist}}. \quad (2)$$

Equation 2 is the general localization-success criterion, with r_k obtained from Eq. 1. Although τ_{dist} can represent a metric radius in geographic coordinates, the VPAIR protocol used here applies an index-based tolerance. A query is counted as correct when the retrieved reference lies within a $\pm R$ -frame window centered on the ground-truth sequence index.

To expose the segmentation model to controlled domain shifts, GCWA introduces a stochastic guided transformation $\mathcal{G}$. Given an original image I , its building mask M , a weather prompt P_w , and a random-noise realization η , the operator synthesizes a candidate in weather domain $w \in \{\text{fog}, \text{snow}, \text{night}\}$:

$$I_w = \mathcal{G}(I, M, P_w, \eta). \quad (3)$$

In Eq. 3, the mask acts as a spatial constraint, the prompt defines the requested appearance domain, and η represents generation stochasticity. A candidate is retained only when it satisfies three complementary conditions: building-region agreement, edge-structure similarity, and a sufficiently strong photometric shift.

First, geometric agreement is measured by the IoU between the inherited mask and the prediction of a fixed reference segmenter:

$$\text{IoU}(M, \text{Seg}(I_w; \theta_{\text{ref}})) \geq \tau_{\text{IoU}}. \quad (4)$$

Here, $\text{Seg}(\cdot; \theta_{\text{ref}})$ is the building-segmentation function parameterized by the frozen reference weights θ_{ref} . The reference model is the YOLOv11 segmenter from our previous study [7], trained only on the original Building Segmentation Dataset and never updated with the synthetic weather domains evaluated during filtering. This separation limits circular validation: generated samples are screened by a model that has not learned from those samples. The threshold τ_{IoU} controls the minimum permitted agreement between the inherited annotation and the candidate image.

Second, boundary stability is assessed on image gradients:

$$\text{SSIM}(\nabla I, \nabla I_w) \geq \tau_{\text{edge}}, \quad (5)$$

where ∇ is the gradient operator and τ_{edge} is the minimum permitted SSIM value for the edge maps.

Equation 5 complements region overlap by detecting changes in high-frequency structures such as roof outlines, corners, and building boundaries. Passing this test bounds edge drift under the selected metric, but it is not interpreted as proof of exact pixelwise invariance.

Third, a candidate must exhibit a nontrivial appearance change:

$$\mathcal{D}_{\text{shift}}(I, I_w) \geq \tau_{\text{shift}}, \quad (6)$$

where $\mathcal{D}_{\text{shift}}$ is a perceptual-difference function and τ_{shift} is its minimum accepted value.

Equation 6 rejects candidates that preserve geometry but fail to express the intended weather shift. We implement $\mathcal{D}_{\text{shift}}$ as the learned perceptual image patch similarity (LPIPS) metric with an AlexNet backbone and 256×256 RGB inputs. LPIPS is comparatively tolerant of small local noise while remaining responsive to changes in illumination, contrast, and texture. The selected value $\tau_{\text{shift}} = 0.10$ excludes near-copies of the source without excessively reducing the accepted set; downstream segmentation and retrieval experiments determine whether the retained transformations improve robustness.

The accepted original and synthetic samples are used to optimize the YOLOv11 segmentation model through the multi-component objective

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{seg}} + \lambda_2 \mathcal{L}_{\text{box}} + \lambda_3 \mathcal{L}_{\text{dfl}} + \lambda_4 \mathcal{L}_{\text{cls}} \quad (7)$$

In Eq. 7, $\mathcal{L}_{\text{seg}}$ penalizes mask disagreement, $\mathcal{L}_{\text{box}}$ penalizes object-localization error, $\mathcal{L}_{\text{dfl}}$ is the distribution focal loss used to refine boundary localization, and $\mathcal{L}_{\text{cls}}$ is the classification loss for the single building class. The coefficients $\lambda_1, \dots, \lambda_4$ balance the gradients contributed by these tasks. Although this objective is not a dedicated metric-learning loss, the shared backbone is trained to preserve building semantics across the verified weather domains; its interme-

diate representation is subsequently used as the descriptor in Eq. 1.

3.2 GCWA Method

To mitigate segmentation degradation under adverse weather, we developed the GCWA method, which is based on an iterative “Generate–Verify–Filter” cycle. Unlike standard image stylization techniques, GCWA turns augmentation into a guided data curation process in which each synthetic sample is checked for preservation of geometric accuracy. Figure 1 shows the overall scheme of the GCWA method.

In the first phase of GCWA, the guided transformation operator $\mathcal{G}$ is implemented using the Gemini 2.5 Flash Image [28] (sometimes referred to as “Nano Banana”) model in image-to-image mode, where the input image serves as a basis for photometric transformation into the target weather domain while preserving scene geometry. To ensure reproducibility, we provide below the configuration of generative-model calls and the prompt structure.

Generation was performed via the Google Gemini API provider (Google AI for Developers / Google AI Studio), specifying the model identifier gemini-2.5-flash-image and a fixed “model/version” string returned by the API response: “models/gemini-2.5-flash-image”.

The model parameters were fixed for all runs:

- temperature: 1.0;
- topP: 0.95;
- topK: 64;
- candidateCount: 1;
- aspect ratio: 1:1;
- MIME type: “image/png”.

To ensure stable results, fixed prompt templates were used, with only the domain variable being changed, $w \in \{fog, snow, night\}$. Table 3 provides an example of the prompt structure (presented as a template and reproducible verbatim).

Each generated candidate I_w undergoes comprehensive consistency analysis with respect to the original mask M through three independent control channels.

- Mask structural-consistency analysis. The masks of the original and generated images are compared according to Eq. 4. This ensures that the main building regions remain in their original locations.
- Edge-similarity verification. Because IoU is robust to minor distortions, we additionally compute the structural similarity index (SSIM) between the gradient maps of the original and generated images. This makes it possi-

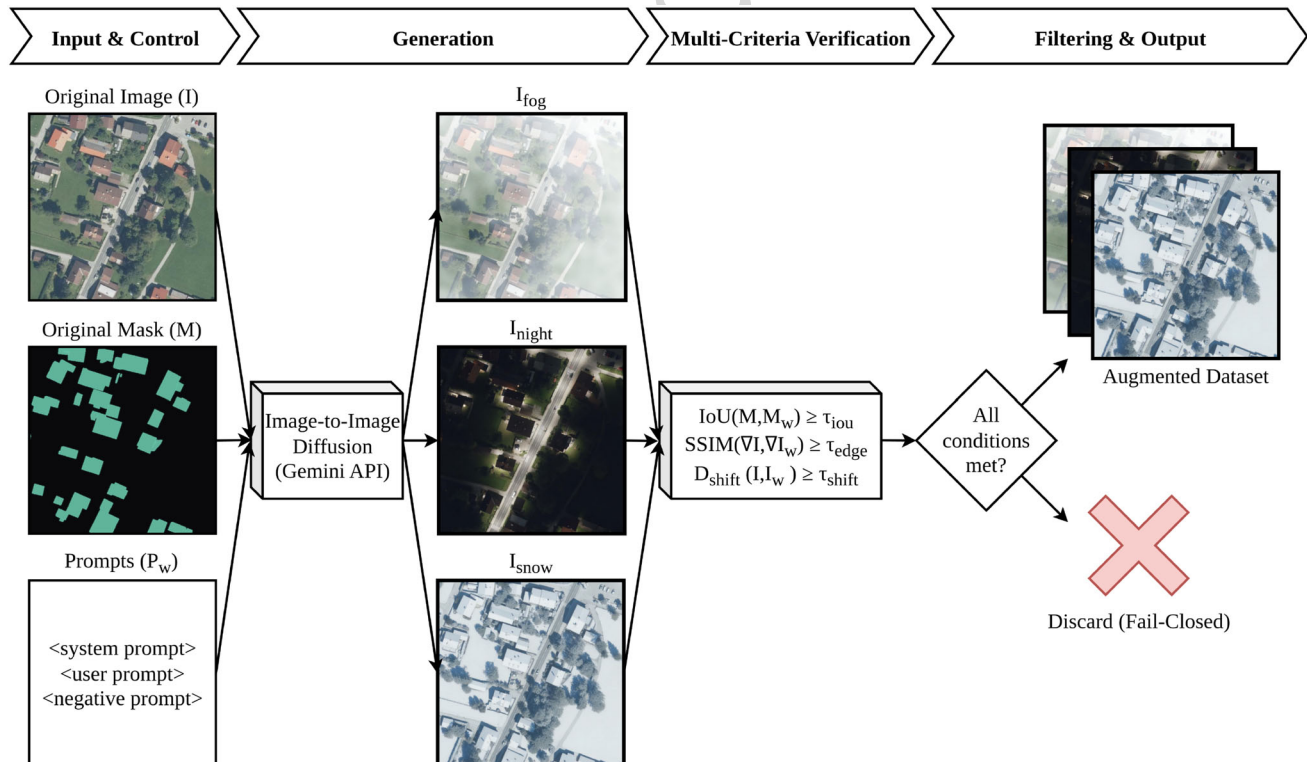


Fig. 1 General scheme of GCWA based on the Generate–Verify–Filter (GVF) pipeline. The method screens the topological integrity of building footprints across multiple domains (fog, snow, night) by applying explicit geometric and structural criteria before dataset expansion

Table 3 Prompt configuration for Geo-consistent Weather Augmentation

Prompt type	Content
System Prompt	You are an image-to-image model. Do not alter geometry or scene layout—no warping, scaling, cropping, rotation, perspective changes. Do not add/remove/move any objects. Preserve all building footprints, roof outlines, edges, corners, and road edges exactly. Return the image at the exact same resolution.
User Prompt	Transform the image to <DOMAIN> conditions: <DOMAIN_DESCRIPTION>.
Negative Prompt	geometry change, warped buildings, shifted edges, new buildings, removed buildings, extra structures, changed roof shape, altered road layout, text, logos, artifacts, blurry edges, duplicated objects, deformation

ble to detect micro-deformations of building lines that can adversely affect descriptor precision.

- Minimum weather-shift strength control (threshold τ_{shift}). Additionally, we assess whether the generated frame differs sufficiently from the original for the augmentation to be meaningful for learning invariance. To this end, we compute the perceptual difference between the original and the candidate ($\mathcal{D}(\cdot)$, in this work – LPIPS). If $\mathcal{D}_{\text{shift}} < \tau_{\text{shift}}$, the candidate is rejected as nearly devoid of weather perturbations.

GCWA follows a fail-closed principle: if none of the generated variants meets the verification thresholds, the image is excluded from augmentation. This is crucial to prevent label noise from entering the training set.

For each accepted sample, metadata are recorded (weather type, verification scores, model version), ensuring full reproducibility and enabling ablation studies of the impact of each weather condition on final VPR accuracy.

To preserve methodological rigor, augmentation is performed independently for each data split (train/test). This guarantees that the model never observes a weather-augmented version of a test image during training, ensuring an objective robustness evaluation under domain shift.

3.3 GCWA Algorithm

Algorithm 1 operationalizes the method for an original dataset $\mathcal{S}$, a weather-domain set $\mathcal{W}$, fixed prompts, verification thresholds, and K generated candidates per source-domain pair.

Line 6 applies Eq. 3 with a candidate-specific noise realization η_k , allowing multiple visual variants when $K > 1$. At

Algorithm 1 GCWA.

Require: Original dataset $\mathcal{S} = \{(I_i, M_i)\}_{i=1}^N$; weather domains $\mathcal{W}$; prompts $\{P_w\}$; thresholds τ_{IoU} , τ_{edge} , τ_{shift} ; candidates per domain K .

Ensure: Augmented dataset $\mathcal{S}_{\text{aug}}$.

```

1:  $\mathcal{S}_{\text{aug}} \leftarrow \mathcal{S}$  ▷ retain all original samples
2: for all  $(I, M) \in \mathcal{S}$  do
3:   for all  $w \in \mathcal{W}$  do
4:      $\mathcal{C} \leftarrow \emptyset$  ▷ valid candidates for domain  $w$ 
5:     for  $k = 1$  to  $K$  do
6:        $I_w^{(k)} \leftarrow \mathcal{G}(I, M, P_w, \eta_k)$  ▷ Generate; Eq 3
7:        $M_w^{(k)} \leftarrow \text{Seg}(I_w^{(k)}; \theta_{\text{ref}})$ 
8:        $s_{\text{IoU}} \leftarrow \text{IoU}(M, M_w^{(k)})$  ▷ Eq. 4
9:        $s_{\text{edge}} \leftarrow \text{SSIM}(\nabla I, \nabla I_w^{(k)})$  ▷ Eq. 5
10:       $d_{\text{shift}} \leftarrow \mathcal{D}_{\text{shift}}(I, I_w^{(k)})$  ▷ Eq. 6
11:      if  $s_{\text{IoU}} \geq \tau_{\text{IoU}}$  and  $s_{\text{edge}} \geq \tau_{\text{edge}}$  and  $d_{\text{shift}} \geq \tau_{\text{shift}}$  then
12:        add  $(I_w^{(k)}, s_{\text{IoU}} + s_{\text{edge}})$  to  $\mathcal{C}$  ▷ Verify
13:      end if
14:    end for
15:    if  $\mathcal{C} \neq \emptyset$  then
16:      select  $I_w^* \in \mathcal{C}$  maximizing  $(s_{\text{IoU}} + s_{\text{edge}})$  ▷ Filter
17:      add  $(I_w^*, M)$  to  $\mathcal{S}_{\text{aug}}$ 
18:    else
19:      continue ▷ fail-closed: discard domain  $w$  for this image
20:    end if
21:  end for
22: end for
23: return  $\mathcal{S}_{\text{aug}}$ 

```

Line 12, candidates that pass all three thresholds are assigned the additive geometric score $s_{\text{IoU}} + s_{\text{edge}}$. Line 16 retains the valid candidate with the strongest combined region and boundary agreement. If the valid set is empty, Line 19 discards that weather augmentation rather than attaching the original mask to an uncertain image. The algorithm always retains the original samples and adds at most one accepted candidate per source and weather domain.

3.4 Datasets

The experiments combine a building-segmentation dataset for learning structural representations with VPAIR for cross-view retrieval. Section 3.4.1 describes the segmentation source data, Section 3.4.2 defines the localization benchmark, and Section 3.4.3 specifies the GCWA-derived domains and leakage-control procedure.

3.4.1 Building Segmentation Dataset

The YOLOv11 model was initialized and fine-tuned on a specialized building-segmentation dataset [40]. It contains 9,665 annotated high-resolution images from heterogeneous urban landscapes and architectural settings: Tyrol (2,999 images), Tripoli (1,078), Kherson (1,053), Donetsk (999), Mykolaiv (739), Kharkiv (602), Mekelle (951), and other cities (1,244). This geographic diversity supports learning building geom-

etry that is not specific to one region. All images and masks were resized to 640×640 pixels for YOLOv11 training and evaluation.

3.4.2 VPAIR Dataset for Localization Evaluation

Localization robustness was evaluated with VPAIR [5], a large-scale aerial VPR and localization benchmark recorded from a light aircraft flying above 300 m with a downward-facing camera. Each query is paired with a high-detail reference rendering, a depth map, and a six-degree-of-freedom (6-DoF) pose. Position is derived from Global Positioning System (GPS) coordinates, and orientation is represented in a local north–east–down (NED) coordinate system. The benchmark covers a trajectory exceeding 100 km and provides 2,706 query–reference pairs together with distractor images.

The present study uses only the urban subset, where buildings and road infrastructure provide the semantic structures required by the proposed descriptor. Accordingly, the reported robustness claims apply to dense built-up scenes; rural, forested, and otherwise weakly structured environments remain outside the validated scope.

3.4.3 Synthetic Weather Domains

GCWA was applied to the Building Segmentation Dataset and to the UAV queries selected from VPAIR. The “fog” domain models reduced contrast, depth-dependent scattering, and attenuation of high-frequency background detail. The “snow” domain introduces snow cover on horizontal surfaces and changes in scene color temperature. The “night” domain reduces global exposure and introduces localized artificial illumination while requiring the original building contours to remain stable. In all three cases, the original satellite reference database is unchanged.

For the cross-domain segmentation experiment, a fixed 80/20 source-level train–test partition was created before augmentation. Every generated counterpart inherited the partition of its source, so neither an original image nor any transformed version could cross the split boundary. Fine-tuning stability was evaluated separately with seven independently randomized 80/20 source-level partitions, each also established before augmentation. The repeated-partition mean and standard deviation are reported in the following subsections; these runs are distinct from the fixed partition used to construct the train-domain–test-domain transfer matrices.

3.5 Experimental Setup

The experimental configuration tests two linked hypotheses: verified GCWA data improve cross-weather building segmentation, and the resulting segmentation features improve

VPR robustness when only UAV queries are weather-perturbed.

3.5.1 Segmentation Model Configuration

Structural features were extracted with the YOLOv11 instance-segmentation architecture [39], specifically the YOLOv11n-seg nano variant, comprising 2,842,803 parameters and 9.7 giga floating-point operations (GFLOPs) at 640×640 input resolution. Each model was fine-tuned for 100 epochs with AdamW, an initial learning rate of 10^{-3} , and weight decay of 0.05. In Eq. 7, the empirical weights were $\lambda_1 = 7.5$ for segmentation, $\lambda_2 = 7.5$ for bounding-box localization, $\lambda_3 = 1.5$ for DFL, and $\lambda_4 = 0.5$ for classification. The two geometry-related components therefore received the largest weights.

3.5.2 GCWA Verification Parameters

Algorithm 1 used $\tau_{\text{IoU}} = 0.85$, $\tau_{\text{edge}} = 0.70$, $\tau_{\text{shift}} = 0.10$, and $K = 1$. The Mask-IoU threshold requires strong overlap between the inherited mask and the fixed reference model’s prediction. The Edge-SSIM threshold requires substantial contour similarity, and the LPIPS threshold rejects transformations that remain visually too close to the source. One candidate was generated per image and weather domain because image generation was the dominant offline cost.

The global operating point was selected on a 1,000-source-image subset. Each source contributed one fog, one snow, and one night candidate. Seven threshold configurations varied one criterion at a time around the selected values while holding the other two fixed. For each configuration, the analysis recorded the proportion of source images with at least one accepted weather candidate, the mean number of valid candidates per source, the retained augmented-set size, and downstream mAP50 under the segmentation protocol in Section 3.6.1. The final configuration was the one with the highest observed mAP50, with acceptance and set size used to characterize the quality–coverage trade-off.

3.5.3 Density-Stratified Threshold-Stability Protocol

The same 1,000-source-image subset was stratified by building-pixel fraction to examine whether the selected Mask-IoU operating point changed with scene complexity. For a source image I with an inherited binary mask $M \in \{0, 1\}^{H \times W}$, building-pixel fraction was defined as

$$\rho(I) = \frac{1}{HW} \sum_{u=1}^H \sum_{v=1}^W M_{uv}. \quad (8)$$

In Eq. 8, H and W are the mask height and width. Images were ordered by $\rho(I)$ and divided into three equal-count

strata, denoted sparse, medium, and dense. Because each source generated one fog, snow, and night candidate, each stratum contained approximately 1,000 image-domain candidates.

Within stratum b , candidate-level acceptance was recomputed from the logged verification scores. Let N_b be the number of candidates in the stratum, and let s_i^{IoU} , s_i^{edge} , and s_i^{shift} be the three scores for candidate i . With $\tau_{\text{edge}} = 0.70$ and $\tau_{\text{shift}} = 0.10$ fixed, acceptance at Mask-IoU threshold τ was

$$\hat{A}_b(\tau) = \frac{1}{N_b} \sum_{i \in b} \mathbb{I} \left[s_i^{\text{IoU}} \geq \tau \wedge s_i^{\text{edge}} \geq 0.70 \wedge s_i^{\text{shift}} \geq 0.10 \right]. \quad (9)$$

Equation 9 uses $\mathbb{I}[\cdot]$ as the indicator function. Mask IoU was examined first on the global 0.05 grid and then on a 0.01 grid over $[0.80, 0.90]$. The operating-point rule used for the pooled analysis was applied separately to the sparse, medium, and dense strata. A descriptive locationwise check repeated the coarse-grid comparison across the seven named source locations; unequal location sample sizes precluded interpreting this comparison as a formal equivalence test.

For descriptive comparison with the observed acceptance gradient, the independent-candidate binomial standard error was calculated as

$$\text{SE}(\hat{A}_b) = \sqrt{\frac{\hat{A}_b(1 - \hat{A}_b)}{N_b}}, \quad (10)$$

Equation 10 provides only a descriptive sampling scale. The three weather candidates generated from one source image may be dependent, so the approximation was not used for formal inference.

3.5.4 Computational Environment

All experiments were executed on Ubuntu using an NVIDIA GeForce RTX 5070 Ti graphics processing unit (GPU) with 16 GB of video random-access memory (VRAM) and driver 580.173.02, an AMD Ryzen 9 9950X central processing unit (CPU) with 16 cores and 32 threads, and 64 GB of random-access memory (RAM). The operating system was Ubuntu 24.04.4 long-term support (LTS). The software stack comprised Python 3.10.12, PyTorch 2.9.0 with Compute Unified Device Architecture (CUDA) 12.8, Ultralytics 8.3.223, Open Source Computer Vision Library (OpenCV) 4.12.0, Numerical Python (NumPy) 2.2.6, scikit-image 0.23.2 for Edge-SSIM, and lpips 0.1.4 for perceptual difference. Gemini 2.5 Flash Image was accessed through the API in batches issued with 16 concurrent requests.

3.5.5 Evaluation Criteria

Building segmentation was measured with mean average precision at an IoU threshold of 0.50 (mAP50) and the F1 score [33]. The F1 score is the harmonic mean of precision and recall and therefore summarizes the balance between false-positive and missed building predictions. Localization was measured with top-1 recall (Recall@1) under the VPAIR tolerance $R = 3$ frames [15]. A query was successful when its highest-ranked satellite match fell within ± 3 sequence indices of the ground-truth reference. The notation $R = 3$ frames is used throughout to distinguish the discrete sequence tolerance from a metric geographic radius.

3.6 Evaluation Setup

Evaluation proceeds in two stages. Section 3.6.1 measures how training-domain composition affects segmentation across original and adverse-weather tests. Section 3.6.2 then measures whether the resulting building-oriented representation improves retrieval when the satellite database remains clean and only UAV queries are perturbed.

3.6.1 Segmentation Evaluation Protocol

The segmentation experiment uses three complementary scenarios. In the baseline scenario, YOLOv11 is trained only on original favorable-condition images and evaluated on the original, fog, snow, and night test domains; this directly measures degradation under domain shift. In the proposed augmented scenario, the model is trained on the combined original and GCWA-accepted domains and evaluated on each individual domain and their union. In the weather-specific scenario, separate models are trained on the original, fog, snow, and night domains and tested in an each-on-each design. The resulting transfer matrices distinguish matched-domain performance from asymmetric cross-domain generalization. All scenarios report mAP50 and F1 under the criteria in Section 3.5.5.

3.6.2 VPR Evaluation Protocol

VPR is evaluated on the urban VPAIR subset described in Section 3.4.2. The satellite database contains only clean reference images. UAV queries are evaluated in the original condition and after accepted GCWA transformations, thereby isolating adverse weather on the airborne observation while keeping the reference map fixed. For every query, the 256-dimensional YOLO descriptor is extracted, Euclidean distances to all satellite descriptors are computed, and the top-ranked index is selected by Eq. 1. Correctness is then determined by Eq. 2 instantiated with the $R = 3$ -frame criterion from Section 3.5.5.

Table 4 Sensitivity of GCWA to verification thresholds τ .

Bold values highlight the selected row's Edge-SSIM and LPIPS thresholds and its resulting acceptance, yield, size, and mAP50

τ_{IoU}	τ_{edge}	τ_{shift}	Acceptance%	Avg valid candidates	Augmented size	mAP50
0.8	0.7	0.1	99	2.25	2253	0.778
0.85	0.7	0.1	96	1.9	1902	0.781
0.9	0.7	0.1	88	1.55	1548	0.778
0.85	0.6	0.1	99	2.35	2354	0.778
0.85	0.8	0.1	86	1.45	1447	0.777
0.85	0.7	0.05	98	2.05	2051	0.779
0.85	0.7	0.15	88	1.65	1646	0.776

The proposed YOLOv11+GCWA representation is compared with CosPlace [41], a convolutional global-descriptor baseline without explicit semantic anchoring; MixVPR [42], a strong holistic descriptor based on spatial and channel feature mixing; and YOLO without GCWA [7], which uses the same segmentation-oriented representation without geometrically verified weather training. This design separates the contribution of GCWA from the underlying YOLO representation and compares the structurally anchored descriptor with established holistic VPR alternatives.

4 Results

This section reports the empirical results obtained with the GCWA procedure defined in Section 3.2 and Algorithm 1. The results cover threshold selection and stability, cross-domain segmentation, repeated-partition fine-tuning, VPR on VPAIR, and computational cost.

4.1 Sensitivity Analysis of GCWA Thresholds

The verification operating point was evaluated on the 1,000-source-image subset. For each source image, one candidate was generated in each of the fog, snow, and night domains, corresponding to the setting $K = 1$. Candidate acceptance therefore required simultaneous satisfaction of the Mask-IoU, Edge-SSIM, and perceptual-shift criteria in Eqs. 4–6. The analysis first varied the three global thresholds and then examined whether the selected Mask-IoU value remained stable across building-density strata.

Table 5 Candidate-level verification by building-density tercile (1,000-image subset, three weather domains)

Density tercile	Acceptance at $\tau_{IoU} = 0.85$ (%)	Coarse-grid optimum	Fine-grid optimum
Sparse	58	0.85	0.83
Medium	62	0.85	0.85
Dense	69	0.85	0.85
Pooled	63	0.85	0.85

4.1.1 Global Threshold Sensitivity and Operating-Point Selection

Table 4 summarizes seven configurations. Acceptance% is the proportion of source images with at least one accepted weather candidate; Avg valid candidates is the mean number of accepted image-domain candidates per source. Augmented size gives the retained synthetic samples, and mAP50 follows the segmentation protocol in Section 3.6.1.

With $\tau_{edge} = 0.70$ and $\tau_{shift} = 0.10$ fixed, decreasing τ_{IoU} from 0.85 to 0.80 increased acceptance from 96% to 99% and augmented size from 1,902 to 2,253, while mAP50 decreased from 0.781 to 0.778. Increasing τ_{IoU} to 0.90 reduced acceptance to 88% and the set to 1,548 samples, again with mAP50 of 0.778.

At $\tau_{IoU} = 0.85$ and $\tau_{shift} = 0.10$, relaxing τ_{edge} to 0.60 produced 2,354 samples at 99% acceptance and 0.778 mAP50. Tightening it to 0.80 produced 1,447 samples at 86% acceptance and 0.777 mAP50. Thus, a 907-sample change corresponded to a 0.001 difference in mAP50.

At fixed geometric thresholds, lowering τ_{shift} to 0.05 produced 98% acceptance, 2,051 samples, and 0.779 mAP50; increasing it to 0.15 produced 88% acceptance, 1,646 samples, and 0.776 mAP50. Across all configurations, mAP50 ranged from 0.776 to 0.781 while augmented size ranged from 1,447 to 2,354. The maximum, 0.781, occurred at $\tau_{IoU} = 0.85$, $\tau_{edge} = 0.70$, and $\tau_{shift} = 0.10$, which was retained thereafter.

4.1.2 Threshold Stability Across Building Density

The density-stratified analysis used the building-pixel fraction $\rho(I)$ defined in Eq. 8 and the candidate-level acceptance

function in Eq. 9. Table 5 reports the selected Mask-IoU thresholds and the acceptance rates obtained with $\tau_{\text{edge}} = 0.70$ and $\tau_{\text{shift}} = 0.10$. Each density tercile contained approximately 1,000 image-domain candidates. These candidate-level percentages use a different denominator from the source-level any-domain Acceptance% in Table 4; the pooled value of 63% and the source-level value of 96% therefore quantify different events.

On the 0.05 grid used for the global sweep in Section 4.1.1, $\tau_{\text{IoU}} = 0.85$ was selected for the sparse, medium, dense, and pooled candidate sets. On the finer 0.01 grid over $[0.80, 0.90]$, the selected value remained 0.85 for the medium, dense, and pooled sets but shifted to 0.83 for sparse scenes. The coarse-grid operating point was also unchanged in the descriptive locationwise check across the seven source locations. Because location-specific sample counts were unbalanced, the locationwise result is reported descriptively rather than as an equivalence test.

At $\tau_{\text{IoU}} = 0.85$, acceptance increased from 58% in sparse scenes to 62% in medium and 69% in dense scenes, giving an 11-percentage-point sparse-to-dense difference. Equation 10 gave an approximate standard error of 1.5 percentage points per stratum. As stated in Section 3.5.3, this descriptive approximation does not model within-source dependence and is not used for formal inference.

Acceptance under the Edge-SSIM and perceptual-shift criteria varied by less than five percentage points among the density strata, whereas the larger gradient occurred in Mask-IoU acceptance. Verification also altered the density composition of the retained synthetic set. Before filtering, each tercile contributed 33.3% of candidates. After filtering, the sparse-scene share decreased to 30.7%, the medium-scene share was 32.8%, and the dense-scene share increased to 36.5%. The maximum post-filter difference from the balanced pre-filter composition was therefore -2.6 percentage

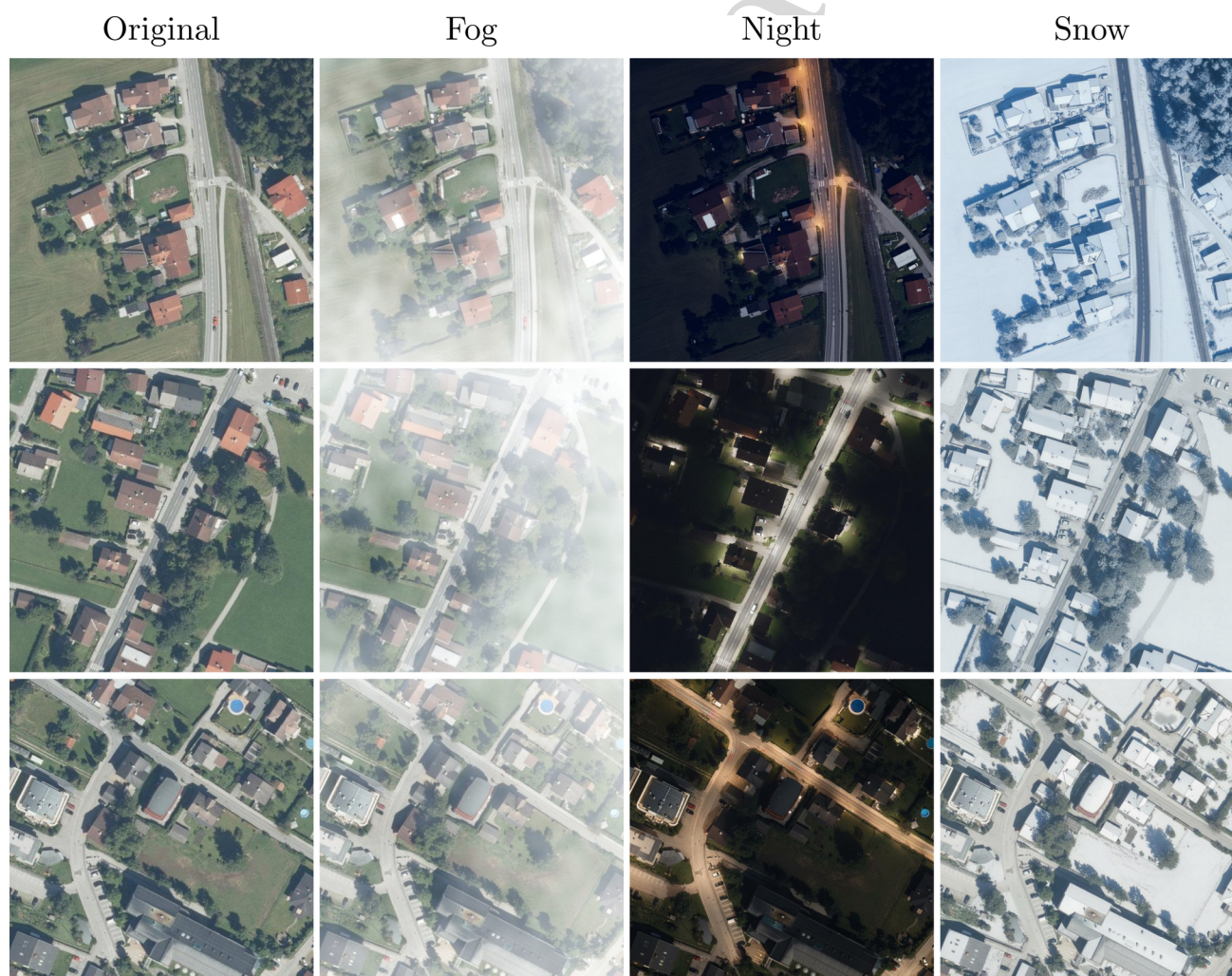


Fig. 2 Visual examples of weather augmentations generated by the GCWA method for three randomly selected samples of the building-segmentation dataset. Columns show, from left to right, the original image, fog, night with artificial lighting, and snow; each row corresponds to one source scene

Table 6 Cross-condition mAP50 matrix for YOLOv11 segmentation models. **Bold** values denote the highest score in each test-domain column

Training Domain	Baseline	Fog	Night	Snow	Augmented
Baseline	0.826	0.397	0.577	0.573	0.590
Fog	0.820	0.816	0.606	0.606	0.714
Night	0.742	0.390	0.700	0.675	0.623
Snow	0.722	0.349	0.529	0.751	0.583
Augmented	0.824	0.815	0.724	0.753	0.781

points for sparse scenes and +3.2 percentage points for dense scenes.

4.2 Cross-Domain Segmentation Evaluation

The cross-domain experiment followed the fixed 80/20 source-level partition described in Section 3.4.3; generated counterparts remained in the same partition as their source images. Four single-domain variants were evaluated: Baseline, Fog, Night, and Snow. The Augmented variant combined all four domains for both training and testing. Figure 2 shows representative source images and their accepted weather-domain transformations. The images retain the spatial layout and building contours required by Eqs. 4 and 5, while the fog, night, and snow columns exhibit the accepted photometric changes required by Eq. 6.

Tables 6 and 7 report the complete train–test transfer matrices. Rows identify the training domain and columns identify the testing domain. Diagonal entries represent matched-domain evaluation; off-diagonal entries quantify direct domain transfer. The Augmented column evaluates each trained model on the union of all four test domains, whereas the Augmented row reports the model trained on the combined domain set.

For matched-domain testing, the highest mAP50 values were 0.826 for Baseline–Baseline, 0.816 for Fog–Fog, 0.700 for Night–Night, and 0.751 for Snow–Snow. The corresponding F1 values were 0.816, 0.811, 0.718, and 0.769. Baseline training showed the largest measured degradation on fog: mAP50 decreased from 0.826 on Baseline to 0.397 on Fog, an absolute reduction of 0.429, and F1 decreased from 0.816 to 0.448, a reduction of 0.368. For Baseline training evaluated on Night and Snow, mAP50 values were 0.577 and 0.573, while F1 values were 0.631 and 0.616.

Table 7 Cross-condition F1 matrix for YOLOv11 segmentation models. **Bold** values denote the highest or tied-highest score in each test-domain column

Training Domain	Baseline	Fog	Night	Snow	Augmented
Baseline	0.816	0.448	0.631	0.616	0.629
Fog	0.814	0.811	0.656	0.655	0.736
Night	0.751	0.441	0.718	0.694	0.660
Snow	0.721	0.383	0.569	0.769	0.618
Augmented	0.815	0.811	0.739	0.772	0.784

Among single-weather variants, Fog training transferred most strongly to Baseline, with 0.820 mAP50 and 0.814 F1; Night and Snow training produced 0.742/0.751 and 0.722/0.721, respectively. On Fog, models trained on Baseline, Night, and Snow obtained mAP50 values of 0.397, 0.390, and 0.349 and F1 values of 0.448, 0.441, and 0.383.

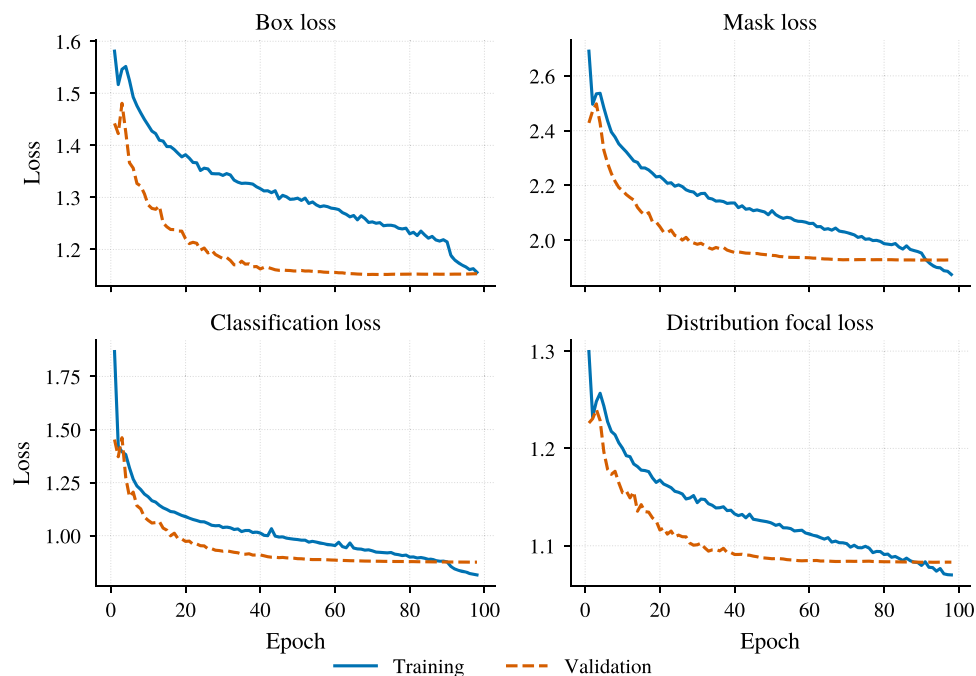
The Augmented model produced mAP50 values of 0.824, 0.815, 0.724, and 0.753 on Baseline, Fog, Night, and Snow, respectively, giving a range of 0.100 across the four individual test domains. Its F1 values were 0.815, 0.811, 0.739, and 0.772, giving a range of 0.076. Relative to Baseline training, the Augmented model changed mAP50 by -0.002 on Baseline, +0.418 on Fog, +0.147 on Night, and +0.180 on Snow. The corresponding F1 changes were -0.001, +0.363, +0.108, and +0.156. On the combined Augmented test set, the Augmented model obtained 0.781 mAP50 and 0.784 F1; the next-highest values were 0.714 mAP50 and 0.736 F1 for Fog training.

4.3 YOLO Fine-Tuning Results

The YOLOv11n-seg configuration specified in Section 3.5.1 was fine-tuned for 100 epochs on the combined original and GCWA-augmented training data. Figure 3 reports the training and validation components of the multi-term objective in Eq. 7. The training curves decreased throughout optimization. Validation losses decreased most rapidly during approximately the first 20–30 epochs and then approached plateaus, with no late increase visible in the reported trajectories.

Figure 4 reports the validation precision, recall, and average-precision trajectories for the bounding-box and mask streams. The mask-stream values are used for the numerical results reported below.

Fig. 3 Training and validation loss curves for the fine-tuned YOLOv11n-seg model. Each panel overlays the training and validation streams for one loss component



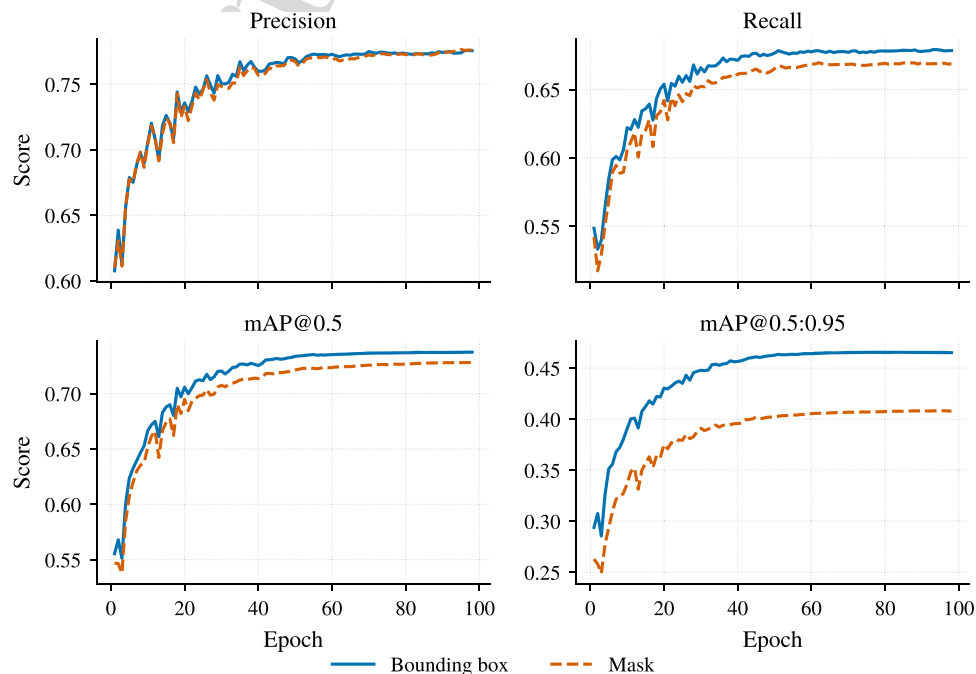
The detection-style confusion matrix in Fig. 5 contains 32,957 correctly identified building fragments, 11,985 false-positive building predictions on background, and 12,312 missed building detections. False negatives exceeded false positives by 327 instances.

Figure 6 presents representative false-positive outputs. Figure 6a contains roof-like geometric patterns in a coastal scene near a road, whereas Fig. 6b contains elongated structures associated with piers, boats, and adjacent coastal infras-

structure. In both cases, the incorrectly segmented regions exhibit local contours or texture arrangements similar to building roofs.

Table 8 reports training and test metrics for the seven independently randomized 80/20 partitions defined in Section 3.4.3. Across the test partitions, mAP50 ranged from 0.722 to 0.727, recall from 0.660 to 0.668, precision from 0.770 to 0.776, and F1 from 0.711 to 0.718. The corresponding mean test values were 0.725 mAP50, 0.664 recall, 0.773

Fig. 4 Validation metrics for the fine-tuned YOLOv11n-seg model, reported separately for the bounding-box (B) and mask (M) prediction streams. Metrics quoted in the text refer to the mask stream



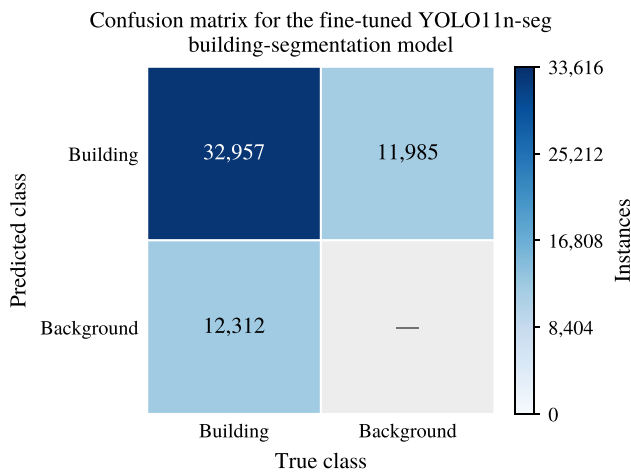


Fig. 5 Confusion matrix for the fine-tuned YOLOv11n-seg building-segmentation model. Rows are predicted classes and columns are ground-truth classes; the bottom right cell is undefined for a detection-style confusion matrix and is left blank

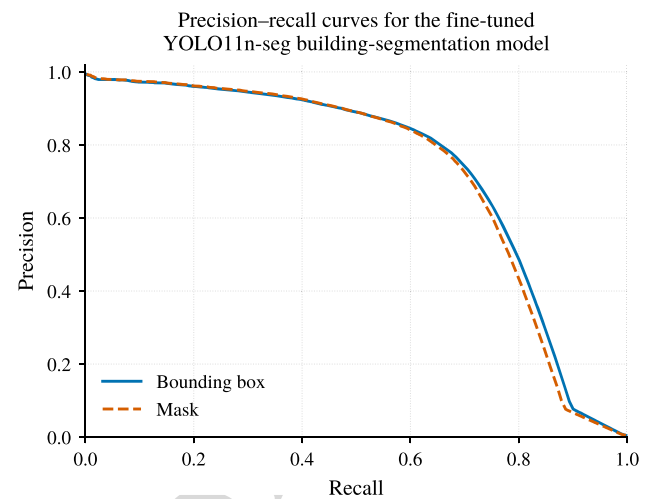


Fig. 7 Precision–recall curves for fine-tuned YOLOv11n-seg; solid and dashed curves denote the bounding-box and mask streams, respectively

precision, and 0.715 F1. Standard deviations were 0.0017, 0.0029, 0.0022, and 0.0026, respectively.

The train–test mean gaps were 0.069 for mAP50, 0.044 for recall, 0.041 for precision, and 0.042 for F1. Partition 3 had the highest test mAP50, 0.727. Its precision–recall curves appear in Fig. 7, where mask AP@0.5 matches Table 8.

4.4 VPR Pipeline Using VPAIR Dataset

The VPR experiment used the urban subset of VPAIR described in Section 3.4.2. The clean satellite database was fixed, and GCWA transformations were applied only to UAV query images, as specified in Section 3.6.2. Descriptor matching followed the nearest-reference rule in Eq. 1; correctness was evaluated with Recall@1 under the $R = 3$ -frame tolerance defined in Section 3.5.5. Table 9 compares the methods introduced in Sections 1 and 3.6.2: CosPlace [41],

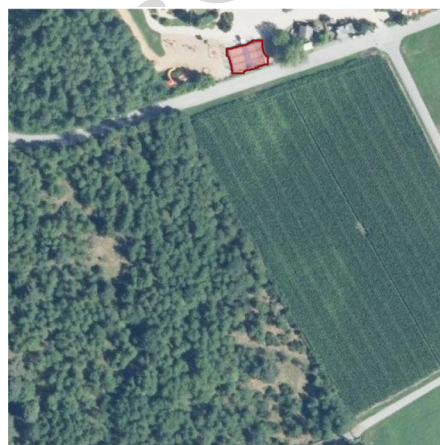
MixVPR [42], YOLO without GCWA [7], and the proposed YOLO with GCWA.

In Original conditions, MixVPR led with 0.210 Recall@1, followed by YOLO with GCWA at 0.195, YOLO without GCWA at 0.181, and CosPlace at 0.140. In Fog, YOLO with GCWA achieved 0.121, exceeding YOLO without GCWA, MixVPR, and CosPlace by 0.042, 0.047, and 0.076.

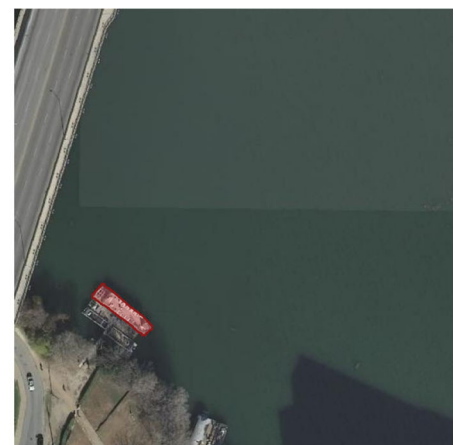
In Night, YOLO with GCWA achieved 0.139, exceeding MixVPR, YOLO without GCWA, and CosPlace by 0.054, 0.056, and 0.088. In Snow, its 0.125 result exceeded the same methods by 0.036, 0.034, and 0.069.

Averaged across Fog, Night, and Snow, adverse-domain Recall@1 was 0.128 for YOLO with GCWA, 0.084 for YOLO without GCWA, 0.083 for MixVPR, and 0.051 for CosPlace. Expressed relative to each method's Original-domain value, the retention rates were 65.8%, 46.6%, 39.4%, and 36.2%, respectively. Because the two YOLO configurations use the same 256-dimensional descriptor construction

Fig. 6 Building-segmentation false positives: (a) roof-like coastal patterns near a road; (b) elongated pier, boat, and coastal-infrastructure structures



(a)



(b)

Table 8 Fine-tuned YOLOv11 metrics across seven random building-segmentation partitions. **Bold** values report test-set mean±SD

Metric	Subset	Random partition							Mean±SD
		1	2	3	4	5	6	7	
mAP50	Train	0.792	0.795	0.796	0.791	0.794	0.793	0.795	0.794±0.0018
	Test	0.723	0.725	0.727	0.722	0.726	0.724	0.725	0.725±0.0017
Recall	Train	0.706	0.709	0.712	0.705	0.710	0.707	0.708	0.708±0.0024
	Test	0.663	0.666	0.668	0.662	0.667	0.660	0.665	0.664±0.0029
Precision	Train	0.812	0.815	0.817	0.811	0.816	0.813	0.814	0.814±0.0022
	Test	0.772	0.775	0.776	0.771	0.774	0.770	0.773	0.773±0.0022
F1	Train	0.755	0.758	0.761	0.754	0.759	0.756	0.757	0.757±0.0023
	Test	0.713	0.716	0.718	0.712	0.717	0.711	0.715	0.715±0.0026

specified in Section 3.1, their 19.2-percentage-point retention difference is the direct measured difference associated with GCWA training under this protocol.

4.5 Computational Cost

The computational measurements are separated into offline dataset construction and online segmentation. Both profiles were obtained on the workstation specified in Section 3.5.4. Offline measurements cover the Generate–Verify–Filter operations in Algorithm 1; online measurements cover the fine-tuned YOLOv11n-seg forward pipeline used before descriptor formation.

4.5.1 Offline Dataset Preparation

Table 10 reports mean per-candidate latency. The Gemini 2.5 Flash Image generation call required 8.0 s on average, with a standard deviation of 2.6 s. The fixed reference-segmentation pass required 6.82 ms, Mask IoU required 0.74 ms, gradient-map construction with Edge-SSIM required 22.20 ms, and LPIPS required 3.31 ms. The verification subtotal was 33.07 ms, yielding a total mean candidate time of 8.033 s. Verification represented 0.41% of total per-candidate time, while generation represented the remaining 99.59%.

Within verification, Edge-SSIM was the largest component at 22.20 ms, followed by reference segmentation at 6.82 ms, LPIPS at 3.31 ms, and Mask IoU at 0.74 ms. The complete verification cascade remained approximately 242 times faster than generation.

Table 9 Recall@1 ($R = 3$ frames) on the urban VPAIR subset. Retention is adverse-domain mean Recall@1 relative to Original. **Bold** values denote the best result in each column.

Method	Original	Fog	Night	Snow	Retention (%)
CosPlace [41]	0.140	0.045	0.051	0.056	36.2
MixVPR [42]	0.210	0.074	0.085	0.089	39.4
YOLO without GCWA [7]	0.181	0.079	0.083	0.091	46.6
YOLO with GCWA (Ours)	0.195	0.121	0.139	0.125	65.8

Table 11 reports full-dataset construction for 9,665 source images and three weather domains. The experiment generated 28,995 candidates and accepted 18,267, corresponding to a pooled candidate-level acceptance rate of 63%. Domain-specific acceptance rates were 61% for Fog, 74% for Snow, and 54% for Night, producing 5,896, 7,152, and 5,219 accepted samples, respectively. Snow yielded 1,256 more accepted samples than Fog and 1,933 more than Night; Fog yielded 677 more than Night.

With 16 concurrent requests, each domain required 1.35 h and the three-domain run required 4.04 h. Sequential execution of the same 28,995 generation calls would require 64.7 h. The 28,995 local verification passes contributed approximately 16 min in aggregate. API expenditure was USD 396 per domain and USD 1,188 overall. Because each domain had the same number of generation calls but different acceptance rates, cost per accepted sample ranged from USD 0.055 for Snow to USD 0.076 for Night; Fog required USD 0.067 per accepted sample, and the pooled value was USD 0.065. At the observed aggregate rate, processing 1,000 source images across all three domains corresponded to approximately 25 min of wall-clock time and USD 123 in API charges. These monetary values describe the provider charge applied during the experiment and exclude tax, storage, and transfer costs.

4.5.2 Online Inference Latency

Online latency was measured on 200 randomly sampled test images at batch size 1 and FP32 precision, with 25 warm-up passes discarded. Image decoding was excluded. Each image

Table 10 Mean per-candidate cost of the GCWA Generate–Verify–Filter loop. **Bold** rows report the verification subtotal and total candidate latency

Stage	Cost
Generation call (Gemini 2.5 Flash Image)	8.0 s $\pm$ 2.6 s
Reference segmentation forward pass	6.82 ms
Mask IoU	0.74 ms
Gradient map and Edge-SSIM	22.20 ms
LPIPS (AlexNet, 256 $\times$ 256)	3.31 ms
Verification subtotal	33.07 ms
Total per candidate	8.033 s
Verification share of total	0.41%

was timed three times and its median was retained, so the dispersion in Table 12 describes image-to-image variation under the stated protocol.

Mean preprocessing, inference, and post-processing times were 0.55, 2.35, and 3.64 ms. Post-processing accounted for most of the 3.18-ms total dispersion. Median latency was 5.88 ms and the 95th percentile was 12.22 ms. The model processed 16.8 ± 17.7 detections per image, sustained 153.0 FPS, and required 307 MB of peak tensor allocation.

The same timing protocol applied to the training split produced 6.22 ± 3.30 ms, 0.32 ms below the test-split mean; the combined standard error was also 0.32 ms. Relative to a 10-Hz update interval of 100 ms, the measured test-split segmentation stage occupied 6.5% of the interval at the mean and 12.2% at the 95th percentile. These values cover preprocessing, network inference, and segmentation post-processing on the reference RTX 5070 Ti workstation; they exclude image acquisition, descriptor construction, database search, temporal filtering, state estimation, memory transfer, power behavior, and control execution.

5 Discussion

5.1 Interpretation of Results and Weather Invariance

The results support the central hypothesis stated in Section 1: synthetic adverse-weather imagery benefits UAV localization

Table 12 Online latency of the fine-tuned YOLOv11n-seg segmenter (200 images; batch size 1; FP32; 640 $\times$ 640). **Bold** values report mean end-to-end latency

Quantity	Value
Parameters	2,842,803
Complexity	9.7 GFLOPs
Preprocessing	0.55 $\pm$ 0.19 ms
Inference	2.35 $\pm$ 0.20 ms
Post-processing	3.64 $\pm$ 3.09 ms
Total latency (mean)	6.54 $\pm$ 3.18 ms
Median latency	5.88 ms
95th-percentile latency	12.22 ms
Sustained throughput	153.0 FPS
Detections per image	16.8 $\pm$ 17.7
Peak tensor allocation	307 MB

when it introduces substantial photometric variation without invalidating geometric supervision inherited from the source image. GCWA implements this principle through the Generate–Verify–Filter procedure in Algorithm 1 and Fig. 1. Mask IoU evaluates agreement between the inherited annotation and the building mask predicted from the transformed image, Edge-SSIM measures contour preservation, and the perceptual-shift criterion rejects candidates with insufficient weather change. Their conjunction addresses a limitation of conventional augmentation [8] and generative augmentation [9]: visual plausibility alone does not ensure that roofs, footprints, and object boundaries remain suitable for dense supervision.

The sensitivity analysis in Section 4 and Table 4 shows that augmentation yield is not an adequate optimization objective. The selected thresholds produced the highest mAP50 of 0.781, 96% source-level acceptance, and 1.9 accepted candidates per source image. Relaxing Mask IoU or Edge-SSIM increased the augmented set to 2,253 or 2,354 samples, respectively, but reduced mAP50 to 0.778, whereas stricter settings reduced training volume without improving accuracy. The selected operating point therefore balances weather diversity against measured label consistency rather than maximizing either sample count or conservatism.

Table 11 Offline dataset-preparation cost per weather domain (9,665 source images; concurrency 16)

	Fog	Snow	Night	All
Candidates generated	9,665	9,665	9,665	28,995
Acceptance rate (%)	61	74	54	63
Accepted samples	5,896	7,152	5,219	18,267
Wall-clock (h)	1.35	1.35	1.35	4.04
API cost (USD)	396	396	396	1,188
Cost per accepted sample (USD)	0.067	0.055	0.076	0.065

The density-stratified findings in Section 4 and Table 5 qualify this conclusion. All terciles retained a coarse-grid selected threshold of 0.85, while the fine sweep placed the sparse-scene value at 0.83 and retained 0.85 for medium and dense scenes. The global setting is therefore stable at the resolution used for model selection, although acceptance remains dependent on scene composition. Candidate-level acceptance increased from 58% in sparse scenes to 69% in dense scenes. This pattern is consistent with a small-union effect in Mask IoU: an absolute boundary discrepancy occupies a larger proportion of a small building-mask union than of a large one. The resulting filter modestly shifts the retained augmented distribution toward dense scenes. This dependence does not invalidate the fixed threshold used in the reported experiments, but it motivates prospectively calibrated alternatives for more heterogeneous corpora.

The transfer matrices in Tables 6 and 7 show that the benefit extends beyond the sensitivity subset. Baseline-only training achieved 0.826 mAP50 on the baseline domain but only 0.397 in fog. The mixed-domain model achieved mAP50 values of 0.824, 0.815, 0.724, and 0.753 on baseline, fog, night, and snow, with corresponding F1 scores of 0.815, 0.811, 0.739, and 0.772. The near-preservation of baseline performance indicates that robustness was not obtained by sacrificing clean-condition accuracy. Instead, the verified samples in Fig. 2 appear to regularize the segmenter toward building shape and layout across photometric changes. This complements cross-view relation learning [4] and simulated-image localization [13], while addressing the earlier problem of synthetic-supervision validity.

The localization results in Table 9 connect segmentation robustness to VPR. YOLOv11+GCWA achieved Recall@1 values of 0.121, 0.139, and 0.125 in fog, night, and snow, compared with 0.079, 0.083, and 0.091 for the same 256-dimensional descriptor trained without GCWA. Because architecture and dimensionality are unchanged, the increase in adverse-weather retention from 46.6% to 65.8% isolates the contribution of geo-consistent training. MixVPR remained strongest in the Original domain at 0.210 versus 0.195 for YOLOv11+GCWA, but retained only 39.4% of its clean-condition Recall@1 across adverse domains. The result is therefore a robustness trade-off rather than universal superiority: holistic appearance cues remain effective under closely matched conditions, whereas the building-based representation degrades less under the tested shifts. The modest absolute Recall@1 values remain consistent with the difficulty of VPAIR [5] and show that GCWA improves ranking without solving aerial localization as a whole.

5.2 Comparative Positioning and Practical Trade-offs

Table 2 clarifies how GCWA differs from the approaches reviewed in Section 2. CosPlace and MixVPR produce holistic global descriptors [41, 42]. Patch-NetVLAD uses multi-scale patches and spatial verification [43]; SegVLAD retrieves semantic segments and their neighborhoods [46]; AnyLoc aggregates self-supervised foundation-model features across diverse environments [44]; and SALAD applies optimal-transport assignment with a dustbin for uninformative features [45]. These methods introduce structure, semantics, or stronger aggregation mainly during descriptor formation or inference. GCWA instead constrains training-data construction. The directions are complementary: verified weather samples could support a structured retriever, while stronger matching could improve the current nearest-neighbor stage.

A direct ranking against SegVLAD, Patch-NetVLAD, AnyLoc, or SALAD is unsupported because they were not executed under the same GCWA protocol. The present evidence isolates GCWA within a matched YOLOv11 descriptor family and compares that pipeline empirically with CosPlace and MixVPR. Resource differences nevertheless explain the deployment choice. The proposed segmenter has 2.84 million parameters, requires 9.7 GFLOPs at 640×640 resolution, and produces a 256-dimensional descriptor of approximately 1 KB per image. The AnyLoc configuration summarized in Section 2 uses a larger foundation-model backbone and a 49,152-dimensional descriptor of approximately 192 KB. For 10,000 reference tiles, the corresponding descriptor storage is roughly 10 MB versus 1.9 GB before index overhead. These values motivate a compact CNN pipeline but do not establish superior accuracy.

Industrial and subterranean parallels reinforce the rationale for structural verification while defining its limits. Tunnel robots require reliable sensing and communication for maintenance decisions [16]. Cable-tunnel imagery is degraded by low illumination, dust, noise, dispersed light, and texture loss, motivating enhancement that recovers inspection detail [17]. Blade inspection and collaborative aerial surface inspection likewise require spatially meaningful evidence rather than merely plausible imagery [18, 19, 37]. These studies support the principle that photometric transformation should preserve task-relevant structure. They are not aerial-VPR baselines, however, and cannot justify transferring the present Mask-IoU or Edge-SSIM thresholds to close-range mechanical inspection without independent calibration.

GCWA provides an auditable filter, modularity with respect to the retriever, and a compact online representation. Its fixed reference segmenter also reduces circularity because the verifier does not adapt to its accepted outputs. The limitations are equally important. Verification depends on reference-segmenter quality; passing the criteria bounds measured disagreement but does not prove exact topology. The building-only prior discards roads, vegetation, waterways, and infrastructure that may aid localization. The cloud generator also introduces provider dependence, rate limits, cost, version drift, and restricted reproducibility.

Section 4 quantifies these trade-offs. Table 10 reports 8.0 s of generation per candidate and 33.07 ms for the full verification cascade, so verification accounts for only 0.41% of the 8.033-s total. Table 11 shows that three weather domains for 9,665 sources required 28,995 generation calls and yielded 18,267 accepted samples. With 16 concurrent requests, the run required 4.04 h and USD 1,189; sequential execution would have required 64.7 h. Dataset construction is therefore feasible at the reported scale, but provider throughput and cost constrain larger corpora.

The online profile in Table 12 is more favorable. YOLOv11n-seg averaged 6.54 ± 3.18 ms per image, with a 5.88-ms median, 12.22-ms 95th percentile, and 153-FPS throughput on the reference workstation. At 10 Hz, segmentation consumed 6.5% of a 100-ms cycle on average. However, the RTX 5070 Ti is a desktop GPU, and the measurement excludes acquisition, descriptor formation, database search, temporal filtering, state estimation, memory transfer, power, and control. The result establishes network-level feasibility on the tested workstation, not complete onboard viability.

5.3 Segmentation Reliability and Interpretation of Model Errors

The seven randomized 80/20 partitions in Table 8 show low within-corpus variation: test mAP50 was 0.725 ± 0.0017 , precision was 0.773 ± 0.0022 , recall was 0.664 ± 0.0029 , and F1 was 0.715 ± 0.0026 . The losses in Fig. 3 and mask metrics in Fig. 4 are consistent with stable fine-tuning. They do not demonstrate geographic generalization because all partitions originate from the same source corpus and augmentation process. The small dispersion therefore indicates robustness to partition choice, not external reproducibility.

Figure 5 reports 32,957 correctly identified building fragments, 12,312 false positives, and 11,985 missed detections. Similar false-positive and false-negative counts indicate no strong bias toward aggressive detection or conservative rejection. The corrected mask AP@0.5 value of 0.727 in Fig. 7 agrees with the best split in Table 8 and permits mission-specific confidence adjustment. Figure 6 shows that boats, piers, coastal infrastructure, roof-like textures, thin structures, and low-contrast roofs can be confused with buildings.

This semantic aliasing explains why improved segmentation does not guarantee correct retrieval. Additional classes, boundary-aware losses, multi-scale processing, or area and compactness filters may reduce these errors, but they require evaluation at both mask and localization levels.

5.4 Limitations and Future Research

The principal limitation is external validity. The experiments use one building-segmentation corpus, one urban VPAIR subset, one cloud generator, three synthetic weather domains, and a fixed clean satellite database. No independently collected flight sequence in real fog, snowfall, rain, or nighttime illumination was evaluated. Future work should freeze all GCWA settings before testing real georeferenced flights and report Recall@1, sequence-level recall, localization-error distributions, confidence intervals, and failures stratified by visibility and building density. A separate reference-side experiment should introduce seasonal or outdated satellite imagery instead of perturbing only UAV queries.

The second limitation is semantic and geographic scope. Buildings are persistent urban landmarks and have supported segmentation-based localization [6, 7], but they are insufficient in farmland, forests, coastlines, and many subterranean environments. Future models should incorporate roads, waterways, vegetation boundaries, and infrastructure and ablate each class. Industrial and tunnel analogies should be converted into controlled experiments with task-, scale-, and sensor-specific calibration rather than used to justify direct threshold transfer.

The third limitation is selection bias from fixed thresholds and $K = 1$. Sparse scenes are rejected more often, and failed generations have no recovery path. Three adaptive strategies merit prospective evaluation without affecting the present results. A quantile-based strategy could choose a separate Mask-IOU cutoff within each calibration stratum to maintain a specified acceptance fraction. A complexity-conditioned strategy could relax the cutoff modestly for sparse scenes and tighten it for dense scenes according to normalized building density. A self-consistency strategy could compare weather-image agreement with the reference segmenter's agreement on the corresponding unmodified image, thereby measuring transformation-induced degradation relative to baseline difficulty. These designs must be calibrated on a separate partition, fixed before testing, and evaluated for downstream accuracy and changes in density, weather, and geographic composition. Increasing K should be assessed jointly with cost per accepted sample, and generator dependence should be tested with a locally deployable model using archived checkpoints, prompts, seeds, and metadata.

The remaining questions concern retrieval design and systems integration. A factorial benchmark should combine GCWA with SegVLAD and resource-matched Patch-

NetVLAD, AnyLoc, and SALAD configurations to separate augmentation, backbone, aggregation, and inference-time geometry. The complete camera-to-pose pipeline must then be profiled on the intended flight computer under realistic power and thermal constraints, including acquisition, segmentation, descriptor construction, indexed retrieval, temporal filtering, state estimation, and control deadlines. Until these evaluations are completed, GCWA should be regarded as a verified synthetic-data curation method with demonstrated urban benchmark gains, not a fully validated safety-critical UAV localization system.

6 Conclusion

Overall, this study demonstrates that geo-consistent verification can transform synthetic weather augmentation into an effective strategy for improving UAV visual localization robustness without sacrificing clean-condition performance. The principal findings can be summarized as follows.

- GCWA converts weather synthesis into a fail-closed Generate–Verify–Filter procedure. Mask IoU, EdgeSSIM, and perceptual-shift criteria prevent a photometrically plausible but geometrically inconsistent image from inheriting the original segmentation label.
- Geo-consistent mixed-domain training mitigates the strongest segmentation failure: fog-domain mAP50 increases from 0.397 for baseline training to 0.815, while the augmented model maintains mAP50 values of 0.724–0.824 across the reported test domains.
- On the urban VPAIR subset, YOLOv11+GCWA obtains Recall@1 values of 0.121, 0.139, and 0.125 in fog, night, and snow. Its adverse-weather retention is 65.8%, compared with 46.6% for the same descriptor without GCWA and 39.4% for MixVPR; MixVPR remains higher in the clean domain (0.210 versus 0.195).
- Verification adds only 33.07 ms to an 8.033-s mean generation cycle, and the segmenter requires 6.54 ± 3.18 ms on the reference desktop GPU. These measurements quantify feasibility on the tested hardware but do not establish embedded end-to-end onboard performance.

The evidence therefore supports GCWA as a robust data-curation method for synthetic adverse-weather training in urban, building-based UAV localization. Its broader validity remains limited by synthetic-only weather evaluation, one generative provider, fixed thresholds, a clean reference database, and the absence of a real embedded flight test.

Future research will be devoted to independent real-weather validation, semantics-driven VPR comparisons, and

complete camera-to-pose profiling on the target flight computer.

Author Contributions All authors contributed to the study conception and design. Volodymyr Vozniak and Oleksander Barmak conceived and conducted the experiments, Volodymyr Vozniak, Oleksander Barmak, and Pavlo Radiuk validated the results, Volodymyr Vozniak, Oleksander Barmak, and Iurii Krak made formal analysis, Volodymyr Vozniak did investigation, Pavlo Radiuk and Iurii Krak conducted data curation, Volodymyr Vozniak and Oleksander Barmak prepared the draft of the manuscript, Pavlo Radiuk and Iurii Krak reviewed and edited the manuscript, Volodymyr Vozniak and Pavlo Radiuk made visualizations, Iurii Krak and Oleksander Barmak provided the supervision. All authors read and approved the final version of the manuscript.

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Data Availability Generated data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request.

Materials Availability This study utilized commercially available software and open-source datasets. No new physical materials or proprietary hardware were developed.

Code Availability The scripts for GCWA generation, verification, YOLOv11 training, and VPR evaluation are available from the corresponding author upon reasonable request.

Declarations

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